

MODULE 1

EXPONENTIAL FAMILY AND LINEAR MODELS

1.2 – 1.5 EXPONENTIAL FAMILY

AND GENERALIZED LINEAR MODELS

REGRESSION ANALYSIS · DATA SCIENCE ENGINEERING · ITESO
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Why one function fits them all

In statsmodels there is a single function: GLM()

You pass `family=Poisson()`, `family=Binomial()` or `family=Gaussian()` and the same algorithm estimates all three. There is no separate function per distribution. Today we find out why that is possible.

- 1 **Recognize** the algebraic form shared by normal, Poisson and binomial.
- 2 **Derive** that form for each of them — with three algebraic moves.
- 3 **Discover** that the canonical link is not chosen: it falls out of the distribution.
- 4 **Assemble** the three components of a generalized linear model.

MOTIVATION

Where the linear model stops

$$E[Y_i] = \mu_i = x_i^T \beta \quad \text{with} \quad Y_i \sim N(\mu, \sigma^2), \text{ independent}$$

The basis of most analysis of continuous data — birthweight against gestational age, for instance.

We want similar methods in two situations where this model does not apply:

The response is not normal

Response variables may follow other distributions — and may even be categorical rather than continuous.

Response variable CHURN

$Y = 1$ (Customer cancel subscription)

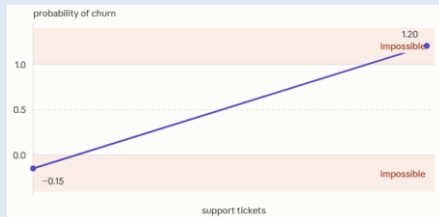
$Y = 0$ (Customer not cancel subscription)

Response variable TICKETS

$Y = 0 - \infty$ (Number of tickets per hour |
Not negative tickets)

The association is not linear

The relationship between response and explanatory variables need not take the simple linear form.



A much wider class of distributions

The properties of the normal distribution are shared by a wider class called the exponential family of distributions.

Normal

Chi-squared

Bernoulli

Wishart

Exponential

Beta

Categorical

Inverse Wishart

Gamma

Dirichlet

Poisson

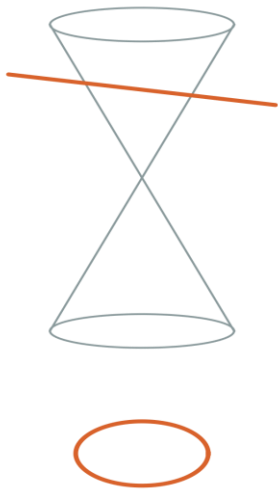
Geometric

The three highlighted ones carry this course: normal for continuous data, Bernoulli for binary, Poisson for counts.

Same idea as conic sections: ellipse, parabola and hyperbola look unrelated until you see they are all cuts of one cone.

One cone, three curves

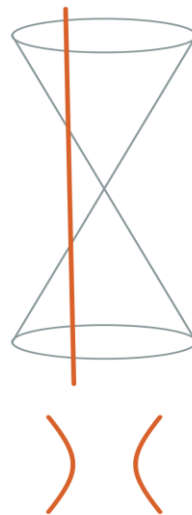
Ellipse



Parabola



Hyperbola



Same cone every time. Only the angle of the plane changes.

The cone is the exponential family. The angle is the distribution you pick — and one theory covers all of them.

DEFINITION

The mold

Let Y be a random variable whose distribution depends on a single parameter θ . It belongs to the exponential family if it can be written as:

$$f(y; \theta) = s(y) t(\theta) e^{[a(y) b(\theta)]}$$

with $s(y) = e^{d(y)}$ and $t(\theta) = e^{c(\theta)}$, this becomes the form we will use:

$$f(y; \theta) = \exp [a(y) b(\theta) + c(\theta) + d(y)]$$

THE TWO THAT MATTER

$$a(y)$$

The only part of the data the parameter ever sees

$$b(\theta)$$

The scale the parameter speaks in — **this is where the link function comes from**

THE OTHER TWO

$c(\theta)$ makes the density integrate to one. $d(y)$ is what is left over. We never look at them again.

Data and parameter meet in exactly one place: $a(y) b(\theta)$. So when you differentiate to estimate θ , $d(y)$ vanishes.

THE RECIPE

Down, sort, back up

Four steps. There is no fifth.

1

Write the density

Straight from the textbook definition

2

Take logs

Products become sums — now you have a list of terms

3

Sort the terms

Ask each one the same question

4

Put it back in the exponential

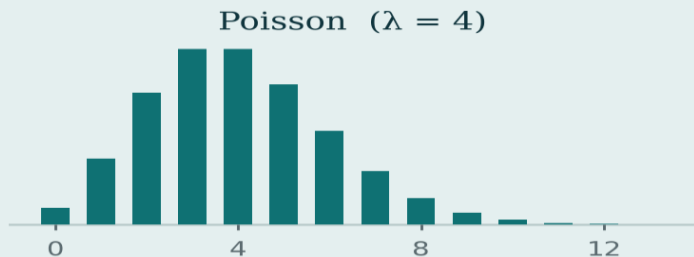
The family form is a density, not a log

If the term has...	it goes to
y and the parameter	$a(y) b(\theta)$
the parameter only	$c(\theta)$
no parameter at all	$d(y)$

Poisson

$$f(y; \theta) = \exp [a(y) b(\theta) + c(\theta) + d(y)]$$

Probability of a number of events in a fixed period, assuming the events occur independently of one another.



Counts: 0, 1, 2, ... · no upper bound · right-skewed · variance = mean

1 WRITE POISSON DENSITY

$$f(y; \lambda) = \frac{\lambda^y e^{-\lambda}}{y!}$$

2 TAKE LOGs

$$\log f = y \log \lambda - \lambda - \log y!$$

3 SORT THERMS

4 BACK INTO THE EXPONENTIAL

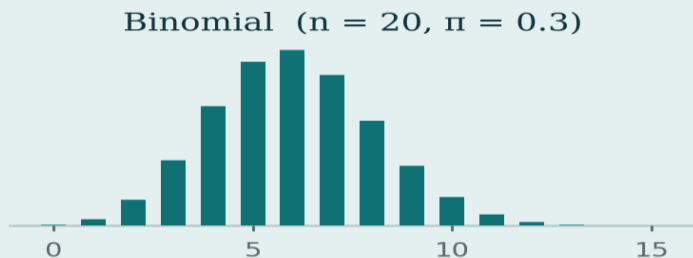
$$f(y; \lambda) = \exp [\underbrace{y}_{a(y)} \underbrace{\log \lambda}_{b(\lambda)} \underbrace{- \lambda}_{c(\lambda)} \underbrace{- \log y!}_{d(y)}]$$

The natural parameter of the Poisson is the logarithm of the mean. That is where the log link of topic 2.4 comes from — nobody chose it.

Binomial

$$f(y; \theta) = \exp [a(y) b(\theta) + c(\theta) + d(y)]$$

Number of successes in n independent trials, each with the same probability π of succeeding.



1 WRITE BINOMIAL DENSITY

$$f(y; \pi) = \binom{n}{y} \pi^y (1 - \pi)^{n-y} \quad \binom{n}{y} = \frac{n!}{y! (n - y)!}$$

2 TAKE LOGS

$$\log f = \log \binom{n}{y} + y \log \pi + (n - y) \log (1 - \pi)$$

3 SORT TERMS

4 BACK INTO THE EXPONENTIAL

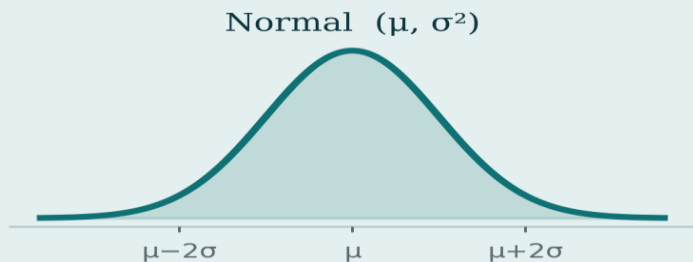
$$f(y; \pi) = \exp [\underbrace{y}_{a(y)} \underbrace{\log \pi - \log (1 - \pi)}_{b(\pi)} + \underbrace{+ n \log (1 - \pi)}_{c(\pi)} + \underbrace{+ \log \binom{n}{y}}_{d(y)}]$$

What just appeared has a name: the logit. Nobody picked it — it fell out of the binomial. That is why logistic regression uses it.

Normal

$$f(y; \theta) = \exp [a(y) b(\theta) + c(\theta) + d(y)]$$

The familiar bell curve: values spread symmetrically around a mean, with a spread that does not depend on where that mean sits.



1 WRITE BINOMIAL DENSITY

$$f(y; \mu) = \frac{1}{(2\pi\sigma^2)^{\frac{1}{2}}} \exp \left[-\frac{(y-\mu)^2}{2\sigma^2} \right]$$

2 TAKE LOGs

$$\log f = -\frac{1}{2} \log (2\pi\sigma^2) - \frac{(y-\mu)^2}{2\sigma^2}$$

3 SORT TERMS

4 BACK INTO THE EXPONENTIAL

$$f(y; \mu) = \exp \left[\underbrace{y}_{a(y)} \underbrace{\frac{\mu}{\sigma^2}}_{b(\mu)} \underbrace{-\frac{y^2}{2\sigma^2}}_{c(\pi)} \underbrace{-\frac{\mu^2}{2\sigma^2} - \frac{1}{2} \log(2\pi\sigma^2)}_{d(y)} \right]$$

The natural parameter is proportional to μ itself. That is exactly why the normal model uses the identity link.

The whole topic is three moves

There are only three algebraic moves in this entire topic

1. Take logarithms 2. Split a product: $(n-y) \log(1-\pi)$ 3. Expand a square: $(y-\mu)^2$

There is no fourth. If you get stuck, it is because there is a product to split or a square to expand.

Distribution	$a(y)$	$b(\theta)$ — natural parameter	Link it induces
Normal	y	μ / σ^2	Identity
Poisson	y	$\log \lambda$	Logarithmic
Binomial	y	$\log [\pi / (1-\pi)]$	Logit

The canonical link is not chosen. It is discovered.

What the natural parameter looks like

Distribution	Real-world parameter	Value	$b(\theta)$
Poisson	$\lambda = 20$ tickets per hour	20	$\log 20 = 3.00$
Binomial	$\pi = 0.30$ open rate	0.30	$\log(0.3/0.7) = -0.85$
Normal	$\mu = 3,000$ g, $\sigma = 500$	3,000	$3000 / 250000 = 0.012$

The right-hand column is the scale the linear predictor lives on

When you fit a Poisson regression, the model does not predict 20 — it predicts 3.00, and then exponentiates. The natural parameter is the scale of the link.

The natural parameter and the canonical link are the same object seen from two sides.

Generalized linear models

Defined over independent random variables $Y_1, \dots, Y_n$, each from the exponential family in canonical form:

$$f(y_i; \theta_i) = \exp [y_i b_i(\theta_i) + c_i(\theta_i) + d_i(y_i)]$$

1

Response variables

$Y_1, \dots, Y_n$, all sharing the same distribution from the exponential family

2

Parameters and design matrix

β and X , where x_i and β are both $p \times 1$ vectors

3

A monotone link function

g , connecting the mean to the linear predictor

$$E[Y_i] = \mu_i$$

$$g(\mu_i) = x_i^T \beta$$

What g is allowed to be

The conditions

- Monotone — so it is invertible
- Differentiable — the estimation algorithm needs g'
- Maps the range of μ onto the whole real line

It may be flat, increasing or decreasing — but it cannot be increasing for some values of μ_i and decreasing for others.

Why the third condition matters

The linear predictor $\mathbf{x}_i^T \boldsymbol{\beta}$ is free — it ranges over all of $\mathbb{R}$. The mean is not: a probability must stay in $(0,1)$, a rate must stay positive.

g stretches the restricted range of μ onto the whole line, so no value of $\boldsymbol{\beta}$ can ever produce an impossible mean.

The link transforms the mean, not the response

$g(\mu_i) = g(E[Y_i]) \neq E[g(Y_i)]$. Fitting a log-link GLM is not the same as taking $\log(y_i)$ and running ordinary regression — and if any $y_i = 0$, the second one does not even exist.

EXAMPLE 1

The normal linear model

$E[Y_i] = \mu_i = \mathbf{x}_i^T \boldsymbol{\beta}$; $Y_i \sim N(\mu, \sigma^2)$, the Y_i independent

The link is the identity

$$g(\mu_i) = \mu_i$$

Because μ already lives on the whole real line, no translation is needed.

Only here can you write

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{e}, \quad \mathbf{e}_i \sim N(0, \sigma^2) \text{ iid}$$

The additive error term is a peculiarity of this case, not a feature of GLMs.